

Lab 1

1. Write a Python program to implement a simple Perceptron with two inputs.

- The program should compute the weighted sum using given inputs, weights, and bias, then apply a step activation function to determine the final output.

```
import numpy as np
# Define inputs and weights
inputs = np.array([1, 0]) # Example input
weights = np.array([0.5, -0.5]) # Initial weights
bias = 0.2 # Bias term
# Compute weighted sum
weighted_sum = np.dot(inputs, weights) + bias
# Apply activation function (Step function)
output = 1 if weighted_sum > 0 else 0
print(f"Perceptron Output: {output}")
```

2. Develop a Python program to implement a Perceptron with three input features.

```
import numpy as np

inputs = np.array([1, 0, 1])
weights = np.array([0.3, -0.2, 0.5])
bias = 0.1

weighted_sum = np.dot(inputs, weights) + bias
output = 1 if weighted_sum > 0 else 0

print(f"Perceptron Output: {output}")
```

3. Implement a Perceptron in Python to simulate the logical AND operation.

- Use appropriate weights and bias so that the Perceptron correctly models the AND gate behavior, then compute the output using a step activation function.

```
import numpy as np

inputs = np.array([1, 1]) # AND gate
weights = np.array([1, 1])
bias = -1.5

weighted_sum = np.dot(inputs, weights) + bias
output = 1 if weighted_sum > 0 else 0

print(f"Perceptron Output: {output}")
```

4. For each of the following scenarios, determine whether the problem is a Classification or a Regression problem.

- i. You are building a system to predict the **number of visitors** to a website tomorrow. Is this a classification or regression problem?
- ii. You are building a system to predict whether an email is **Spam or Not Spam**
- iii. You are working on medical data and want to predict a patient's **blood pressure value**.
- iv. You want to predict whether a tumor is **Benign or Malignant** based on medical images.
- v. Suppose you are working on weather prediction and use a learning algorithm to predict tomorrow's temperature (in degrees Centigrade/Fahrenheit).
- vi. You want to predict the **rating (1 to 5 stars)** a customer will give to a product.
- vii. Suppose you are working on weather prediction, and your weather station makes one of three predictions for each day's weather: Sunny, Cloudy or Rainy. You'd like to use a learning algorithm to predict tomorrow's weather.

5. For each of the following scenarios, identify the type of machine learning approach being used (Supervised Learning, Unsupervised Learning, or Reinforcement Learning).

1. A system learns to play a video game by receiving rewards when it wins and penalties when it loses.
2. A system is trained using labeled images of cats and dogs.
3. A system groups news articles into topics without knowing the topic labels in advance.
4. A model predicts house prices based on historical data where the prices are already known.
5. A robot learns how to walk by continuously adjusting its movements based on feedback from its environment.
6. A system detects unusual credit card transactions without having labeled examples of fraud.
7. A speech recognition system is trained using audio recordings with their correct transcriptions.
8. An online recommendation system improves its suggestions based on user clicks and interactions over time.
9. A dataset of customer information is analyzed to discover hidden customer segments without predefined categories.
10. A model learns to classify emails as spam or not spam using previously labeled emails.